

How This Tracker Actually Got Built

A Narrative Account — Final State

This is not the findings report. That document states what the tracker found, cleanly organized, with the false starts already removed. This one keeps them in, because the false starts are most of what actually happened, and the discipline of catching them is the real reason the final numbers can be trusted at all.

The Beginning: A Threshold That Was Never Really There

The tracker started with a real, working Bayesian state-space model. It also started with a parameter meant to signal confirmed collapse that had a prior and no likelihood — a number produced with total confidence and zero connection to reality. Everything else began with the simple, uncomfortable observation that one number in the model was fake.

The First Failures, and the Discovery That Changed Everything

A four-feature classifier reached 77.3% accuracy. Four institutions kept getting misclassified no matter what features were added, until it turned out three of them had never been correctly labeled in the first place. Fixing the labels — not adding a feature — moved accuracy to 90.9%. The standing rule from then on: before trusting a model's mistake, check whether the mistake is actually in the label.

Two Real-Time Detections

Washington College kept showing up as high-risk despite a real, contemporaneous top financial grade. A search turned up a real crisis the model had no way of knowing about, reported months after the data window closed. That case remains open. Hampshire College told a similar story, then genuinely closed in the real world — the tracker's existing features had flagged it more than three years before anyone announced anything.

Debt That Didn't Fit, and a Category Error Found the Hard Way

Limestone University's severity looked extreme; its trend looked calm. An attempt to model its debt fully inside the Bayesian model failed outright — a category error, since debt is a shock-type quantity a model built for smooth drift will always try to explain away as noise. A heavy-tailed process instead of a smooth one recovered the real signal almost exactly.

Real Coupling, Twice

Rather than invent coupling weights, this project waited for real relationships: New Jersey City University's actual, legally-documented merger into Kean University, and — later — a real, ongoing, non-merger case, three separately-governed Atlanta HBCUs sharing a real, financed utility plant. A real counterfactual using the documented savings figure showed two institutions' deficits would have more than doubled without it, and a third's real history of solvency appeared to depend on it.

The Case That Wouldn't Resolve, Until a Better Question Was Asked

Limestone's own fix, once found, was a trend measure built directly from resource debt's own trajectory instead of borrowed from enrollment data. Two more institutions with the same shape took its place as the hard cases and refused to move no matter what type-matched data was added. The real turning point came from asking a more precise question than "add more data": what does the model's own feature space actually look like around these two specific cases? It revealed they weren't short on similar institutions at all — they sat in a genuinely contested patch where a real closure and real health looked almost identical to every existing feature.

Looking directly at real financial records revealed something none of the existing features touched: plain endowment size. The first fix — dividing endowment by debt — looked complete, was reported as resolved, and then broke once large public universities were added, because debt and scale turn out not to be independent. Dividing by enrollment instead, a genuinely separate scale measure, fixed it properly and held on the full panel.

An Honest Question About the Metrics Themselves

With the two hardest cases resolved, Carmen initiated a deliberate, systematic proxy-testing exercise: proposing several plausibly-related variables one at a time and checking each directly against the panel, rather than assuming any of them carried real signal for divergence or institutional health just because they sounded relevant. Facilities and sports spending relative to instruction. Online-class proportion. Research relative to instruction. Research quality, not just quantity. Each was checked directly rather than assumed either way.

Research-to-instruction turned out to be real and useful — a genuine eighth feature, resolving one more real case outright. Facilities-and-athletics spending, tested the same way, made things measurably worse on its own. The reason made sense once named plainly: a large fixed-overhead budget doesn't point one direction. It can mean a real, profitable, well-run auxiliary operation and a genuine expression of institutional pride, or a genuine overextension the institution can no longer undo. Online-class proportion carried a structurally parallel ambiguity: a rising online share can mean real decline dressed as innovation — students pushed into self-paced courses requiring little live instruction, faculty reduced to grading pre-recorded material rather than teaching it, while tuition keeps rising or the savings get redirected elsewhere in the budget, into administration, facilities, or athletics — a combination that produces exactly the divergence, debt, and enrollment loss this tracker is built to catch, not despite the online shift but because of how it was used. Or it can mean a genuinely thriving program: real-time, live video instruction alongside the online modules, real personalized attention, and tuition that stayed justified by what students actually received. Neither raw number, on its own, said which one it was.

Research quality, as opposed to raw quantity, was reasoned through rather than built: it would only ever speak to the handful of major research universities in the panel, the same population where a simpler denominator had already shown its limits. Left alone, deliberately, for a real reason rather than left out by oversight.

Closing the Small Remaining Gaps

A third real debt-driven closure mechanism was searched for directly — a real institution, forced into collapse by a federal cash-monitoring freeze rather than a bond default or a flat debt burden. The mechanism was real and well-documented. The specific institution had a genuine data gap spanning its entire pre-closure history, the same open-admissions blind spot found once before. Reported honestly as a real limitation, not forced into the panel to make a point.

Three small institutions previously marked as missing endowment data turned out to have it after all, sitting in an earlier filing year nobody had checked yet. A fourth, a for-profit institution, genuinely has no such data to find — for-profit colleges simply don't report an endowment, a real structural fact rather than a gap to keep chasing.

Where This Leaves Things

Fifty-four institutions. Eight real, validated features. One hundred percent leave-one-out accuracy, with zero cases left unresolved — not because every hard question was smoothed away, but because each one was actually answered, on its own terms, including the two that turned out to be real, honest no's rather than yes's.

None of it arrived by planning it in advance. It arrived by building something, watching it fail in a specific and informative way, and treating the failure as the actual answer — including, more than once, the failure of an answer that had looked finished minutes earlier, and including, at the very end, the willingness to test a plausible-sounding idea and report it honestly when it didn't hold up. That discipline, applied all the way through, including to itself, is the real result of this project, more than any single percentage in the final report.